Natural Language Processing

From Fundamentals to Research-Driven

Equipping You with Research Depth and Industry Skills

By:

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What You Will Be Able to Do

- Build **end-to-end NLP pipelines** (from raw text to deployed model)
- Work with **Transformers and LLMs** (BERT, GPT, RAG, PEFT)
- Evaluate models beyond accuracy: **bias, robustness, error analysis**
- Complete a **mini research / paper implementation project** that is GitHub- and CV-worthy
- Be ready for:
  - **Industry roles** in NLP / LLM / AI engineering
  - **Research / PhD** in modern NLP

Stage 1: Foundations & Classical NLP

- **NLP Foundations**
- What is NLP, history, and scope
- Application landscape (search, chatbots, summarization, etc.)
- Tasks taxonomy, datasets, annotation, and challenges (ambiguity, bias, hallucination)
- **Text Processing & Tokenization**
- Cleaning, normalization
- Tokenization strategies, subword methods (BPE, WordPiece, SentencePiece)
- Out-of-vocabulary issues
- **Classical NLP Baselines**
- Bag of Words, TF-IDF
- Naïve Bayes, Logistic Regression for NLP
- **Text Classification & Evaluation**
- Sentiment analysis
- Confusion matrix, Precision / Recall / F1
- Cross-validation, proper evaluation

Stage 1: Why It Matters (Industry + Research)

- After this Stage you can:
- Build a **complete classical sentiment / topic classifier**
- Design a **clean preprocessing + tokenization pipeline**
- Correctly **interpret model metrics** instead of just quoting accuracy
- **Industry:**
  - You can quickly build **baseline NLP systems** for real data (customer reviews, social media, emails)
- **Research:**
  - You understand **strong baselines and evaluation**, which every paper must have

Stage 2: Embeddings & Sequence Modeling

- **Vector Semantics & Embeddings**
- Word2Vec, GloVe
- Limitations of static embeddings and idea of contextual embeddings
- **Sequence Labeling**
- POS tagging, Named Entity Recognition
- BIO tagging, evaluation for sequence tasks
- **Neural NLP Foundations (Applied)**
- Padding, masking, batching sequences
- Loss functions for NLP, overfitting and regularization
- **RNNs in NLP (Applied Recap)**
- Using LSTM/GRU for text classification and NER
- Limitations of RNNs in practice

Stage 2: Why It Matters (Industry + Research)

- Use **embeddings** to improve classical models and deep models
- Build **NER systems** (e.g., extract names, places, orgs from text)
- Implement robust **data loaders and training loops** for NLP in Tensorflow / PyTorch
- **Industry:**
  - Skills for **information extraction from contracts, logs, medical text, financial data**
- **Research:**
  - Ready to work on **representation learning** and **sequence labeling** papers

Stage 3: Transformers, LLM Adaptation

- **Transformers for NLP (Applied)**
- Practical view of self-attention
- Positional encoding, encoder/decoder roles
- HuggingFace ecosystem and pretrained transformers
- **BERT-Style Models**
- Masked Language Modeling
- Fine-tuning pipeline
- Transfer learning in NLP
- **GPT-Style Models**
- Decoder-only models
- Text generation, decoding strategies
- Prompting basics
- **LLM Adaptation Techniques**
- Pretraining vs finetuning
- Prompt engineering
- Evaluation pitfalls for LLMs

Stage 4: Evaluation, Multimodal, Projects

- **Retrieval Augmented Generation (RAG)**
- Chunking documents
- Embedding retrieval and vector databases
- Grounding and hallucination mitigation
- **NLP System Evaluation**
  - Benchmarking
  - Bias and fairness
  - Robustness testing
  - Error analysis
- **Multimodal NLP**
  - Text + image
  - Video + transcript

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Final Research Project

- **Type: Mini research**
- **Paper Implementation**
  - BERT fine-tuning, RAG, summarization, etc.
- **Novel Mini Research**
  - Examples:
    a. Urdu RAG chatbot
    b. Hate speech detection for Pakistani social media
    c. Multimodal video QA system

Final Project Expectation

- **Final Project = Research Paper Implementation**
- Must include:
- Literature review
- Model implementation
- Experimental analysis
- Discussion of results
- This course is your **foundation**, not a coding-only subject.

What You Gain by the End

- **Technical:**
  - End-to-end skills from **classical NLP → Transformers → LLMs → RAG → Multimodal**
- **Industry:**
  - Portfolio of **code, models, and a research-style project**
- **Research:**
  - Ability to **read, implement, and extend modern NLP paper**

Final Message to Students

- “Deep Learning is not about models.
- it is about **thinking, experimenting, and improving systems**.”

Policy and Guidelines

Instructor Information

<table>
<tr><td>Instructor</td><td>Dr. Zohair Ahmed</td><td>E-mail</td><td>zohair.ahmed@isb.nu.edu.pk</td></tr>
<tr><td>Current Position</td><td>Assistant Professor</td><td>Department</td><td>AI & DS</td></tr>
<tr><td colspan="4">PhD (AI/NLP) : Central South University, China.
Research Interest : Natural Language Processing, Multimodal NLP, Sentiment Analysis, Large Language Models, Deep Learning</td></tr>
<tr><td>Office</td><td>203-B, C Block, 2nd Floor</td><td>Tel</td><td></td></tr>
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Prerequisite(s)

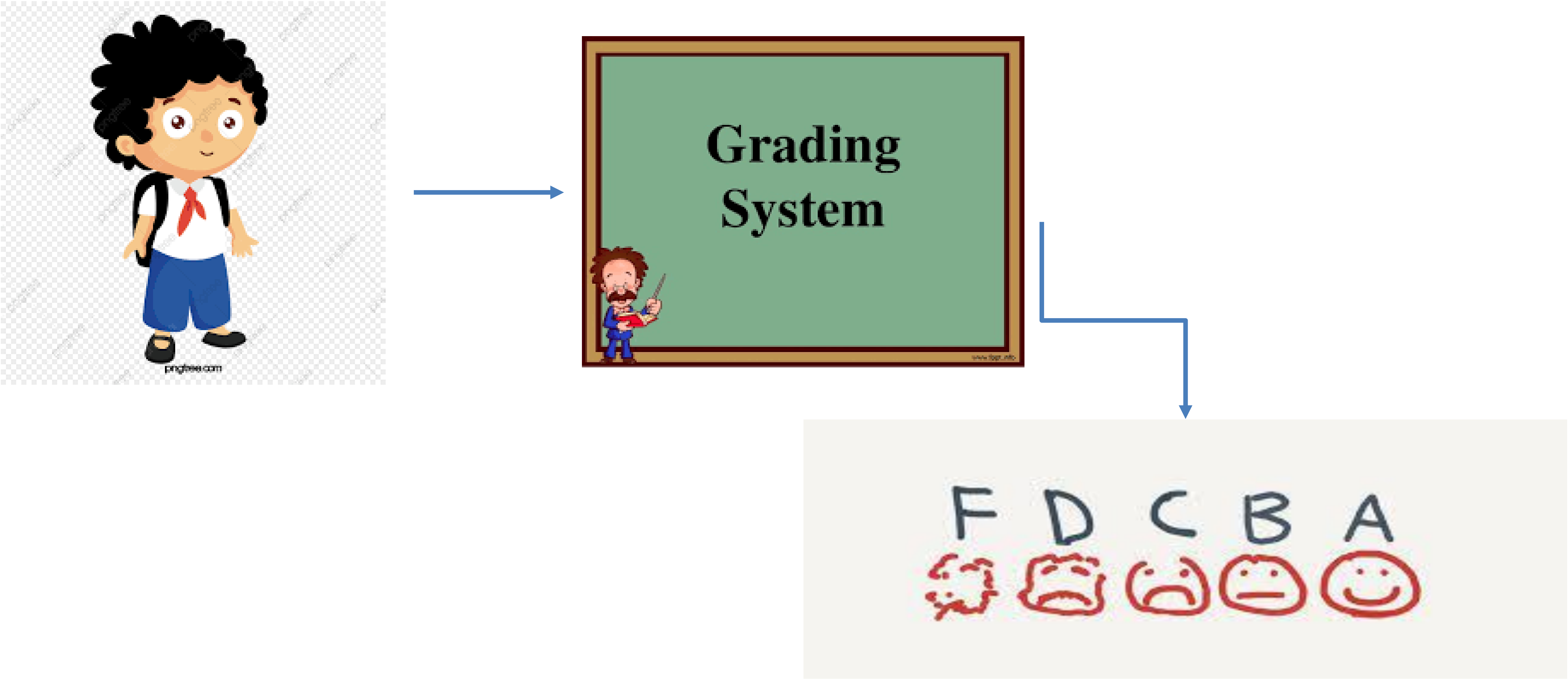
- There is no official prerequisite for this course.
- Concepts of Machine Learning
- Good programming skills especially with one of the following tools:
  - Python / C++/Java
- Machine Learning Frameworks
  - Scikit-Learn
  - **Tensorflow / Keras** , **PyTorch**, DEEPLEARNING4J, Microsoft Cognitive Toolkit, Caffe

Reference Books

- *Jurafsky & Martin – Speech and Language Processing 3rd Edition*

Grading Policy

- Absolute Grading



Explanation of Assessment

- All assignments and Quizzes carry equal weightage
- Enough Time will be given for Assignments for their implementation.
- Quizzes can be announced ~~or unannounced~~.
- Project will be can be done in ~~Groups (max 3) or~~ individually.
  - Project include writing a research paper for publication.
  - You will be using **word / Overleaf for Writing the paper** and presenting the results.

Course Plagiarism Policy

- Plagiarism in any kind of assessment including project or sessional/ final exam, assignments quizzes, will result in F grade in the course.
- So what is it?

> Plagiarism is **presenting someone else's work or ideas as your own, with or without their consent**, by incorporating it into your work without full acknowledgement. All published and unpublished material, whether in manuscript, printed or electronic form, is covered under this definition.

Types of plagiarism

Global plagiarism

Passing off an entire text by someone else as your own

Verbatim plagiarism

Directly copying parts of someone else's work

Paraphrasing plagiarism

Rephrasing someone else's ideas to present them as yours

Patchwork plagiarism

Stitching together parts of different sources

Self-plagiarism

Recycling your own past work

Missed Assessment

- Retake of missed assessment items (other than sessional/ final exam) is NOT allowed.
- Missed assessment item (other than sessional / final exam) will earn zero marks
- Late submission will NOT be accepted.
- For missed sessional/ final exam due procedure will be followed.
- No change is any deadline

Course Plagiarism Policy

- Project Paper
  - Plagiarism for paper will be checked via Turnitin software
  - Any paper with score greater than 20% will be considered plagiarized and ZERO marks will be awarded.
  - Content will be also checked for AI Generated plagiarism, it must not exceed 35% too.

Attendance policy

- Students are supposed to have 100% attendance.
- The minimum attendance requirement at all levels and in all courses is 80%.
- The relaxation of 20% attendance has been given only to cover any planned events or unforeseen situations.
- I will take attendance at the start of lecture, within 10 minutes.
  - Ideal: 0 - 10 minutes Present 10-15 mins marked as Late, beyond 15 you will be **marked ABSENT**.

Student Responsibility

- Mobile Usage
  – Use of mobile phones should be avoided.
- Discussion
  – Class participation is highly appreciated
  – You can ask questions in the class(Highly Appreciated)
  – Discussion among students is not allowed.